INCH-POUND

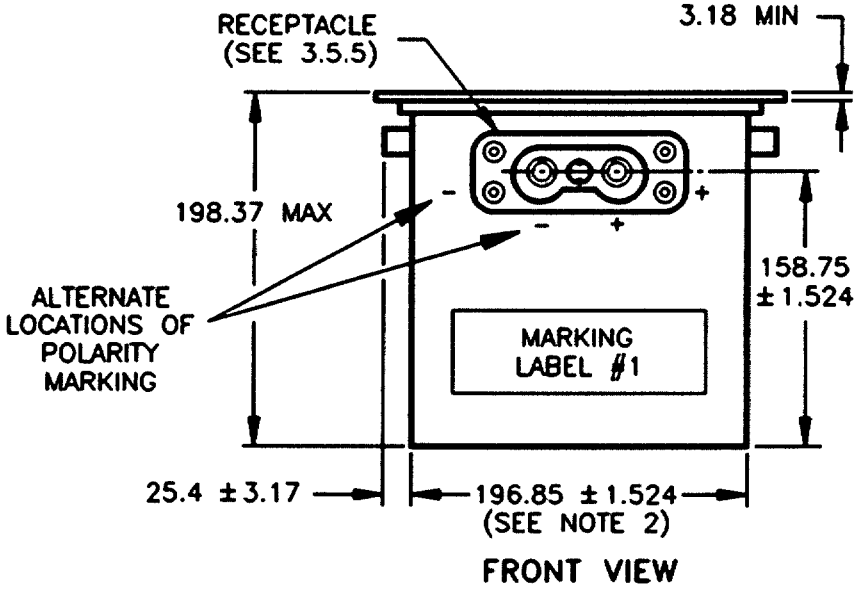
MIL-B-8565/11C
31 MARCH 1993
SUPERSEDING
MIL-B-008565/11B
27 OCTOBER 1992

MILITARY SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MEDIUM-RATE, TYPE 1, MAINTENANCE-FREE, 24-VOLT, 10-AMPERE-HOUR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation:
MIL-B-8565

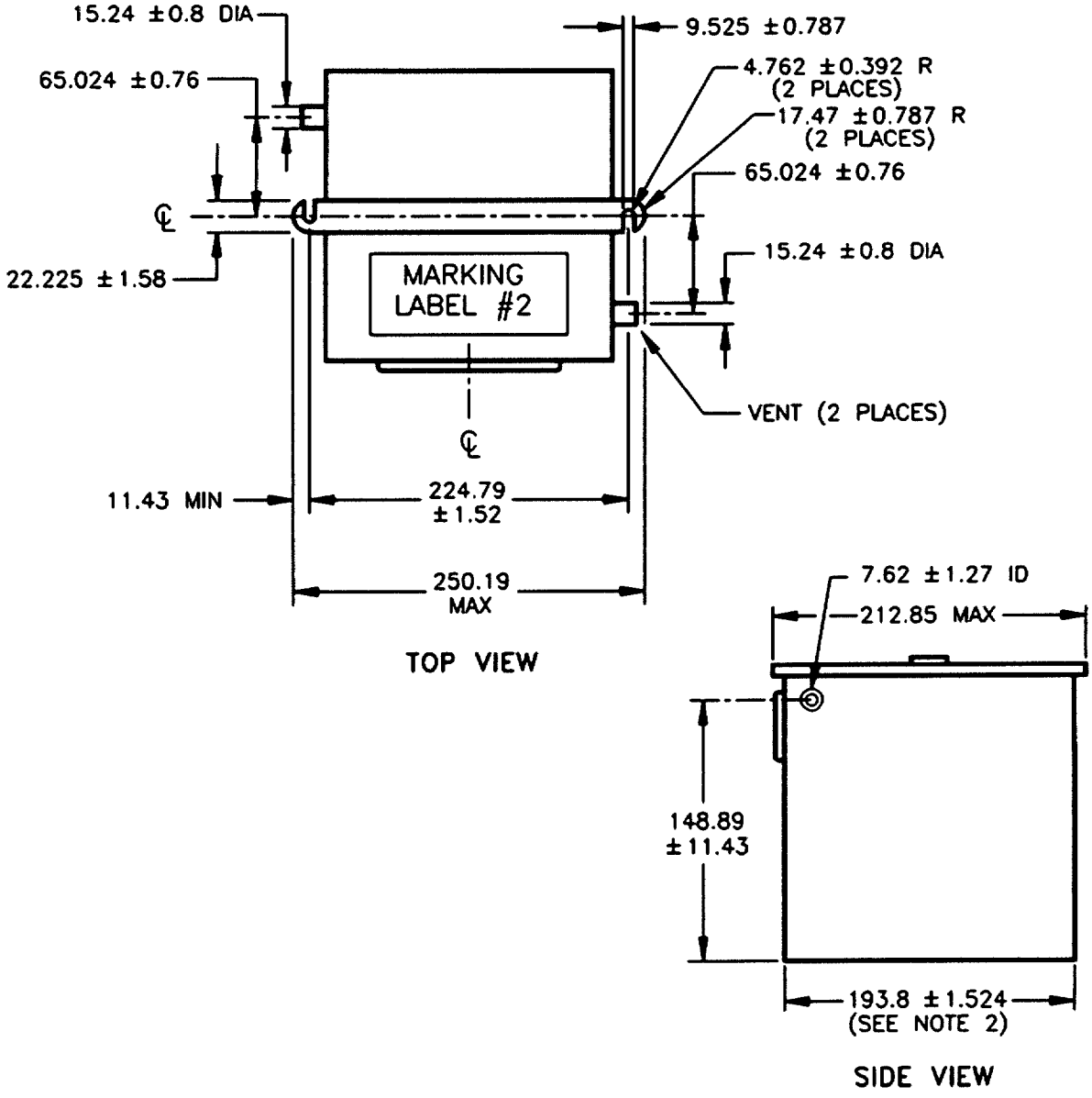


NOTES:
1. All dimensions are in millimeters.
2. Allow additional 10mm width for cover.
3. Abbreviations used in this figure are maximum (MAX), minimum (MIN) and part number (PN).

FIGURE 1. PN D8565/11-1 dimensions.

AMSC N/A FSC 6140
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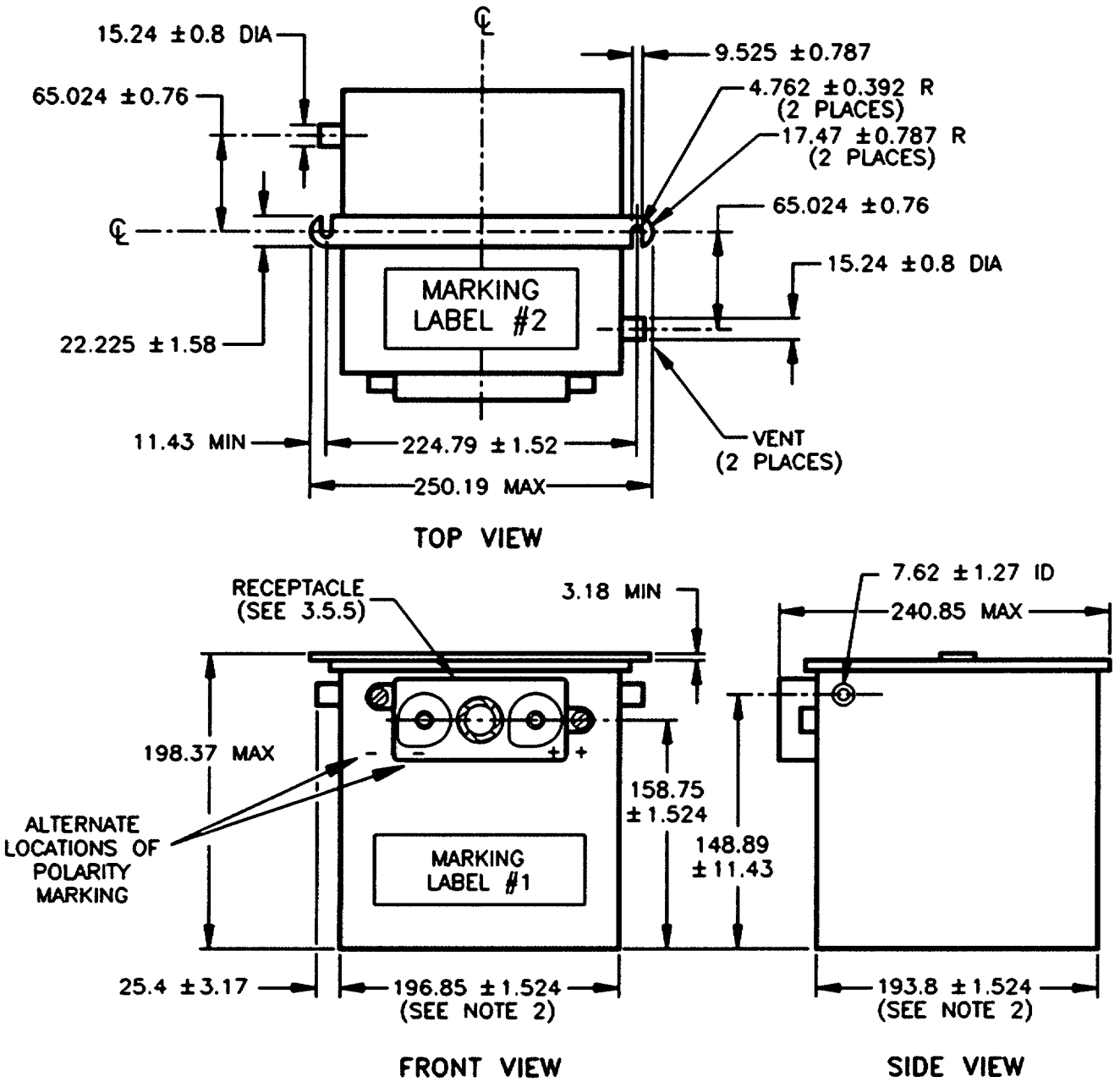
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NOTES:
1. All dimensions are in millimeters.
2. Allow additional 10mm length for cover.
3. Abbreviations used in this figure are diameter (DIA), inside diameter (ID), maximum (MAX), minimum (MIN), radius (R) and part number (PN).

FIGURE 1. PN D8565/11-1 dimensions-Continued.

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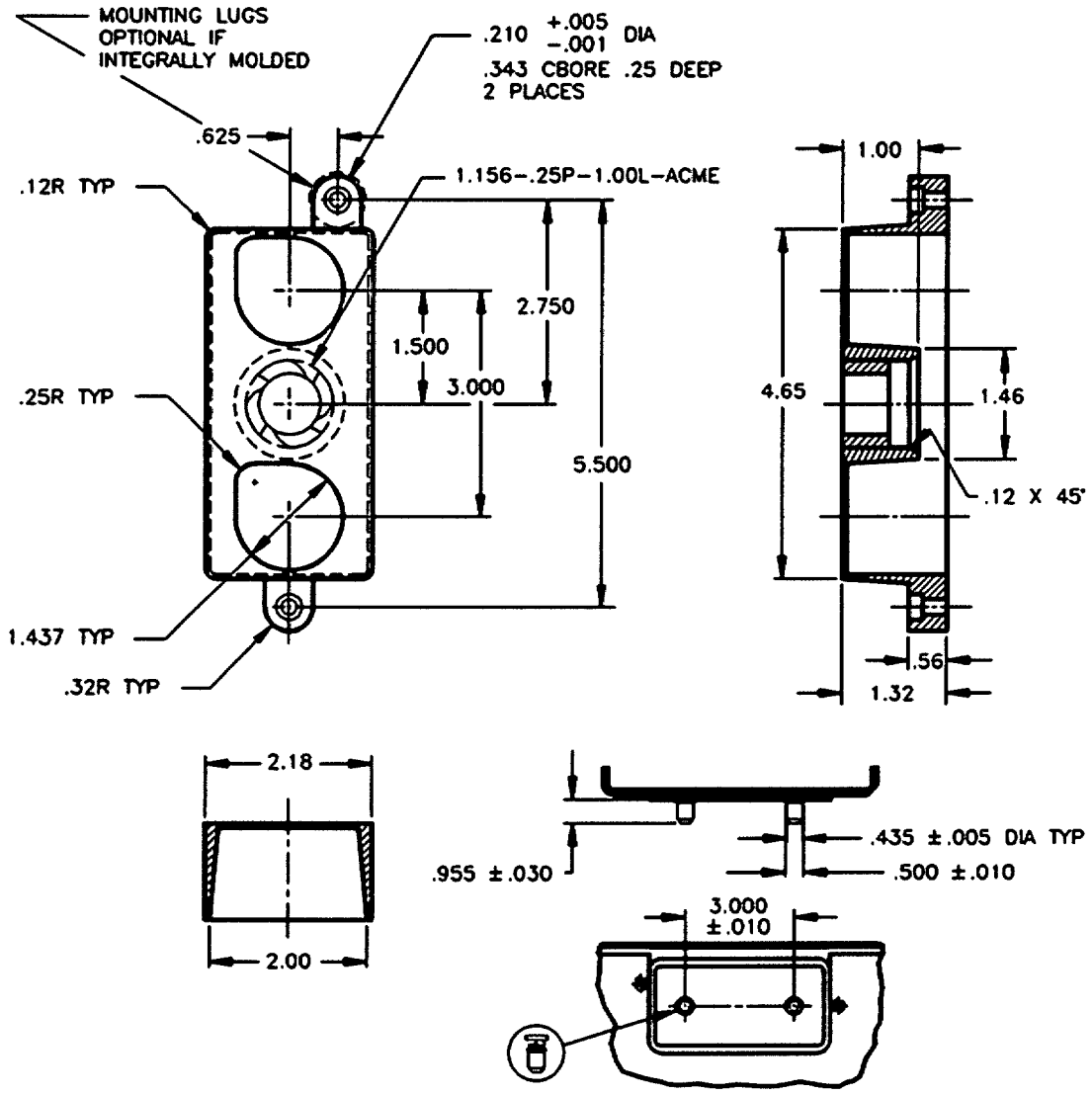


NOTES:

1. All dimensions are in millimeters.
2. Allow additional 10mm to width and length of cover.
3. Abbreviations used in this figure are diameter (DIA), inside diameter (ID), maximum (MAX), minimum (MIN), radius (R) and part number (PN).

FIGURE 2. PN D8565/11-2 dimensions.

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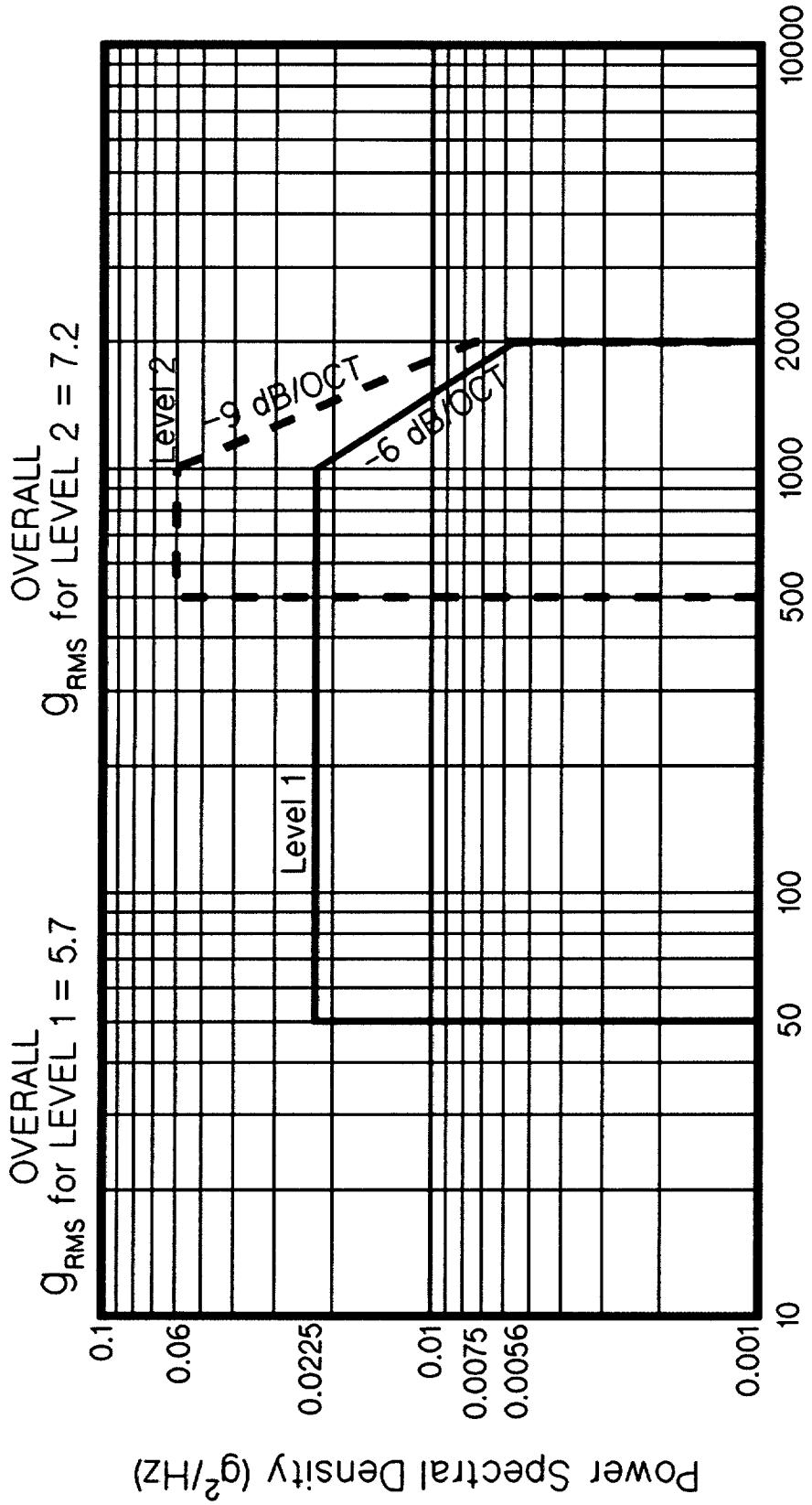


NOTES:

1. Material or finish to be acid resistant.
2. All dimensions are in inches.
3. Tolerances are X.XX (± .03 inches), X.XXX (± .010 inches).
4. Receptacle to case attachment method optional for nonmetallic containers.
5. Draft angles allowed in compliance with commercial die casting practice.
6. Abbreviations used in this figure are counterbore (CBORE), diameter (DIA), radius (R) and typical (TYP).

FIGURE 3. Typical receptacle adapter.

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NOTE:

1. Abbreviations used in this figure are acceleration of gravity (g), decibel (dB), hertz (Hz), octave (OCT) and root mean square (RMS).

FIGURE 4. Random vibration test levels.

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REQUIREMENTS:

1. The dimensions of the battery shall be as shown on figure 1 or figure 2 of the applicable specification sheet.

2. The part number of the batteries shall be D8565/11-1 for figure 1 and D8565/11-2 for figure 2.

3. The rated capacity of the battery shall be 10 ampere-hours, based on its ability to conform to the life test final discharge. The life test final discharge is a 1-hour discharge at 24°C ambient temperature with a 10-ampere load (1C-rate) to a cutoff voltage of 18 volts.

4. The weight of the battery shall be not greater than 15.8 kg.

5. For installation in the aircraft, the battery shall be supported as shown in the conventional mounted example on figure 1 of MIL-B-8565. Battery life in the aircraft shall be 3 years.

6. Table I of the applicable specification sheet lists the quality assurance tests.

7. Delete all requirements for cells and delete all requirements for threaded-post connected batteries.

8. The battery shall conform to MIL-B-8565 for type I, medium rate, except as modified by the following.

3.5.5 Receptacles. Add the following: "For D8565/11-1, the female receptacle (front view) shall be dimensionally equal to MS3509, style 1. For D8565/11-2, the female receptacle shall be dimensionally equal to figure 3 of the applicable specification sheet. Either receptacle may be integrally molded in the container."

3.6.3.2.2 Identification and warranty marking. Delete the following: "Warranty Expiration Date".

3.6.3.2.3 Battery caution marking (type 1 battery only). Delete and substitute the following: "The marking for label 2 shown on figure 1 of the applicable specification sheet shall show the following for battery type D8565/11-1 and D8565/11-2. The lettering height shall be 0.25 inch minimum. The marking shall be in white or black print in accordance with MIL-M-13231.

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MIL-B-8565/11C

MAINTENANCE-FREE BATTERY

DO NOT REMOVE COVER

THIS BATTERY IS TO BE REMOVED FROM
SERVICE 3 YEARS FROM DATE FIRST PLACED
IN SERVICE

CAUTION!

DO NOT USE EXCESSIVE TORQUE

ON HOLD-DOWN STRAP BOLTS"

3.6.8 Capacity and electrical performance. Delete table II and substitute the following:

TABLE II. Capacity performance requirements (type 1 battery).

Requirement number	Discharge cutoff volts & temperature	Rate of discharge factor	Minimum time to cutoff voltage	
(1)	18.0 volts at 24 ± 3°C	1.0C	1 hour	
(2)	18.0 volts at 49 ± 3°C	1.0C	1.1 hours	
(3)	14.4 volts at -12 ± 3°C	6.5C	3 minutes	
(4)	14.4 volts at -30 ± 3°C	6.5C	1 minute	
(5)	14.0 volts at 27 ± 3°C	6.5C	5 minutes	
(6)	10.0 volts at 27 ± 3°C	9.0C 2.5C	1 minute 14 minutes	15 minutes

4.6.11 Constant voltage discharge (14.0 volts). N/A.

4.6.13.1 Cycling test (type 1 battery). Delete "2-hour charge" and substitute "4-hour charge".

4.6.15 Charge and discharge test at low temperature. Delete "-18°C (0°F)" and substitute "-12°C (10°F)". In steps b and d, delete "1.0 C-rate to 18.0V" and substitute "65 amperes to 14.4 volts".

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4.6.20 Special tests. Add the following:

"4.6.20.1 Fifteen-minute discharge at 27°C. The battery shall be charged as specified in accordance with 4.5.1.3 and stored under the specified temperature and storage conditions (see 4.5.1.2). The battery shall then be discharged at a rate specified in accordance with requirement (6) of table II, a 90-ampere discharge for 1 minute followed immediately by a 25-ampere discharge for 14 minutes. The batteries shall maintain a terminal potential of 10.0 volts or greater throughout the total 15-minute duration of the two discharges. The battery shall meet the requirement of 3.6.8.

"4.6.20.2 Three-minute discharge at -12°C. The battery shall be charged as specified in accordance with 4.5.1.3 and stored under the specified temperature and storage conditions (see 4.5.1.2). The battery shall then be discharged at a rate specified in accordance with requirement (3) of table II, a 65-ampere discharge to a terminal potential of 14.4 volts. The battery shall meet the requirement of 3.6.8.

"4.6.20.3 One-minute discharge at -30°C. The battery shall be charged as specified in accordance with 4.5.1.3 and stored under the specified temperature and storage conditions (see 4.5.1.2). The battery shall then be discharged at a rate specified in accordance with requirement (4) of table II, a 65-ampere discharge to a terminal potential of 14.4 volts. The battery shall meet the requirement of 3.6.8."

4.6.23 Vibration. Add the following: "The battery shall also be subjected to two separate random vibration tests along each orthogonal axis. The first test shall be for a period of 20 minutes per axis to level 1 of figure 4 of the applicable specification sheet. The second test shall be for a period of 5 minutes per axis to level 2 of figure 4 of the applicable specification sheet. The battery shall be discharged at a 1-ampere rate during vibration and shall be charged as specified in accordance with 4.5.1.3 prior to beginning each axis of vibration."

9. Revision letters are not used to denote changes due to the extensiveness of the changes.

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TABLE I. Inspection of batteries and order of test.

Order of test	Description of inspection	Requirements paragraph	Inspection method paragraph	Qualification inspection			Quality conformance inspection		
				Qualification sample number			Group A tests	Group B tests	
				1	2	3		1	2
1	Visual & mechanical examination	3.5, 3.6.1 & 3.6.3	4.6.3	X	X	X	X	X	X
2	Dimensions & weight	3.6.5	4.6.4	X	X	X	X	X	
3	Dielectric strength	3.6.4	4.6.5	X	X	X	X	X	X
4	Color & marking	3.6.3	4.6.8	X	X	X	X	X	X
5	Conditioning charge	4.5.1.4	4.6.9	X	X	X		X	X
6	Constant voltage discharge	3.6.8	4.6.11	X	X		X	X	
7	Strength of receptacle	3.6.9	4.6.12			X			X
8	Life cycling	3.6.10	4.6.13	X				X	
9	Capacity discharge at 49°C	3.6.8	4.6.14		X				
10	Charge & discharge test at low temperature	3.6.8	4.6.15		X				X
11	Discharge while inverted	3.5.8 & 3.6.8	4.6.16		X				X
12	Altitude	3.6.11 & 3.6.12	4.6.17		X				X
13	Mechanical shock	3.6.11 & 3.6.12	4.6.18	X					
14	Temperature shock	3.6.11 & 3.6.12	4.6.19	X				X	
15	Fifteen-minute discharge	3.6.8	4.6.20.1		X		X		X
16	Three-minute discharge	3.6.8	4.6.20.2		X				
17	One-minute discharge	3.6.8	4.6.20.3			X			X
18	Temperature rise & float	3.6.13	4.6.21		X				X
19	Battery gas emission test	3.6.14	4.6.22		X				
20	Vibration	3.6.16	4.6.23		X				X
21	Humidity	3.6.17	4.6.24		X				X
22	Salt fog	3.6.18	4.6.25	X				X	
23	Ground storage	3.6.15	4.6.26			X			
24	Physical integrity at high temperature	3.6.12	4.6.27	X					
25	Impact resistance test	3.6.19	4.6.27		1/				
26	Final examination	3.6.4	4.6.30	X	X	X		X	X

1/ For nonmetallic battery container designs, two outer nonmetallic battery containers must be supplied. The containers shall be subjected to the impact resistance test of 4.6.27.

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Navy - AS

Review activity:
Air Force - 99

Preparing activity:
Air Force - 80

Agent:
Navy - AS

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